

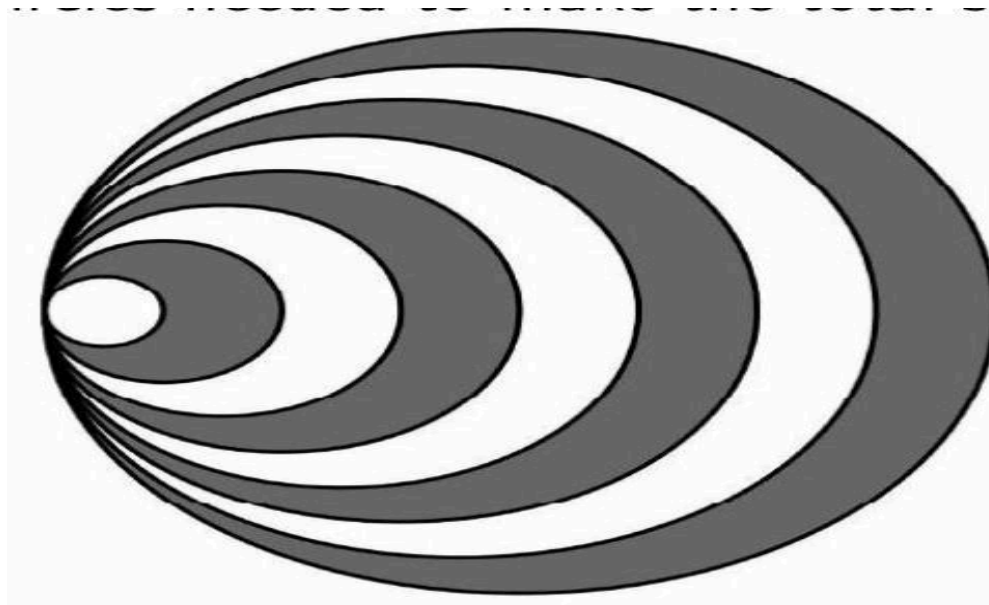
## 2023 AMC 10A Problem 15 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10A Problem 15. Great practice for AMC 10, AMC 12, AIME, and other math contests

### 2023 AMC 10A Problem 15

**Problem:**

An even number of circles are nested, starting with a radius of 1 and increasing by 1 each time, all sharing a common point. The region between every other circle is shaded, starting with the region inside the circle of radius 2 but outside the circle of radius 1. An example showing 8 circles is displayed below. What is the least number of circles needed to make the total shaded area at least  $2023\pi$ ?



**Answer Choices:**

A. 46

B. 48

C. 56

D. 60

E. 64

**Solution:**

Suppose there are  $2n$  circles. The shaded area is then

$$\begin{aligned}
 A &= \pi(2^2 - 1^2) + \pi(4^2 - 3^2) + \cdots + \pi((2n)^2 - (2n-1)^2) \\
 &= 3\pi + 7\pi + \cdots + (4n-1)\pi \\
 &= \pi \sum_{k=1}^n (4k-1) \\
 &= \frac{\pi(3+4n-1)n}{2} \\
 &= n(2n+1)\pi.
 \end{aligned}$$

To find the least value of  $n$  such that  $A \geq 2023\pi$ , note that  $n \approx \sqrt{\frac{2023}{2}} \approx 32$ . Because  $31(2 \cdot 31 + 1) = 1953$  and  $32(2 \cdot 32 + 1) = 2080$ , the required value is  $n = 32$ , and there are  $2 \cdot 32 = (\text{E}) \boxed{64}$  circles.

The problems and solutions on this page are the property of the MAA's [American Mathematics Competitions](#) 